

LECTURE 4.1



Diagnosing Violations

ECONXXXXX Econometrics I

Sam Deegan
University College Dublin
PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

Notes

Weeks already covered are greyed; today's is marked.

What Last Week's Joint Test Assumed

Last week we tested three restrictions at once and read one number off the output.

- What did that number assume about the errors?
- We tried several specifications before we reported one. What did the searching do to the number?
- Collinearity widened a standard error. Did it move the estimate?

Notes

Two minutes with your neighbour. The third question is the shape of today: a fault that costs you precision and leaves the estimate alone.

Lecture Plan

- **Heteroskedasticity.** The estimates survive, the standard errors do not, and that is why robust is the default.
- **Serial Correlation.** The same damage, but read it as a symptom of something left out rather than a disease to treat.
- **Misspecification.** The one that is different in kind, because it takes unbiasedness away.

Notes

Reading: [to settle].

SECTION 1

Heteroskedasticity

1. Heteroskedasticity

2. Serial Correlation

3. Misspecification

What Each Violation Costs: Unbiasedness or Precision

The goal of this section: say what a non-constant error variance costs you, and be exact about what it leaves untouched.

Table 1: Each fault against the promise of week six.

Fault	Still unbiased?	Still best?	Usual SE right?
Heteroskedasticity	Yes	No	No
Serial correlation	Yes	No	No
Wrong shape	No	No	No

Notes

Three rows, and the third one is not a worse version of the first two. It is a different column that has gone.

The Error Fans Out With Schooling

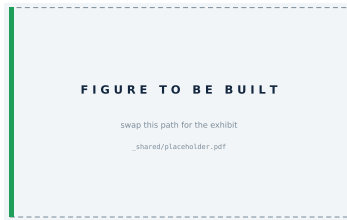


Figure 1: Residuals from the usual fit, plotted against years of schooling. The band is narrow on the left and wide on the right: people with more schooling are spread further from the line.

Notes

Nothing about the line is wrong here. The line is in the right place. What has changed is how far a person is likely to sit from it, and that depends on where they are along the axis.

The Variance of the Slope Under Heteroskedasticity

The question this answers: how far would the slope move across repeated samples when the error variance is not the same for everybody?

$$\text{Var}(\hat{\beta}_1) = \frac{\sum_i (x_i - \bar{x})^2 \sigma_i^2}{[\sum_i (x_i - \bar{x})^2]^2} \quad (1)$$

- $\hat{\beta}_1$: the slope least squares computes from one sample
- x_i : years of schooling for person i
- $\bar{x}$: the sample mean of schooling
- σ_i^2 : the error variance for person i

Set every σ_i^2 equal to the same σ^2 and this collapses to the formula from week 6. That is the only difference, and it is the whole difference.

Estimating the Slope Variance With Squared Residuals

The question this answers: we cannot see σ_i^2 , so what do we put in its place?

$$\widehat{\text{Var}}(\hat{\beta}_1) = \frac{\sum_i (x_i - \bar{x})^2 \hat{u}_i^2}{[\sum_i (x_i - \bar{x})^2]^2} \quad (2)$$

- $\hat{u}_i$: the residual left for person i by the fitted line

Notation: the error and the residual

u_i is the error, which belongs to the population and which nobody ever sees. $\hat{u}_i$ is the residual, which belongs to the fit. The robust standard error is the second standing in for the first, and the hat is carrying all of that.

Why Robust Standard Errors Are the Default Choice

The estimator is due to White (1980), with the same idea reached independently by Eicker (1967) and Huber (1967) a decade earlier. It has been available in every package you will use for forty years.

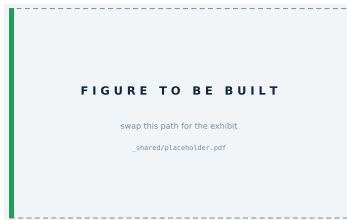


Figure 2: The classical standard error and the robust one for the same slope, as the error is made to fan out more steeply. The estimate barely moves; the two standard errors come apart.

Test first and correct afterwards, and your reported standard error depends on a test you ran and did not report. Use the robust one from the start and it does not.

Arguments For and Against Robust Standard Errors by Default

Angrist & Pischke (2017) argue that most of what an introductory course teaches about inference can be replaced by the instruction to use robust standard errors, and that time spent testing assumptions and applying generalised least squares fix-ups is time badly spent.

The honest qualification

In small samples the robust estimator is itself unreliable, and the usual version is the least reliable of the family. Long & Ervin (2000) and Imbens & Kolesár (2016) both recommend the corrected variants — the ones your software calls HC2 and HC3 — once the sample is small or a few observations carry unusual leverage. MacKinnon & White (1985) is where those variants come from.

Notes

So: robust by default, and the corrected variant by default too. That is a one-word change in your code, not a chapter.

Vote: Which Estimates Are Still Unbiased?

The error fans out with schooling, exactly as in the figure. We fit the usual line and report the usual standard error.

- The slope is biased and the standard error is fine.
- The slope is fine and the standard error is too small.
- The slope is fine and the standard error is wrong, and you cannot say in advance which way.
- Both are wrong.

Notes

Vote, argue with your neighbour, vote again. Then: what evidence would settle it?

Pause

Notes

Two minutes. Write one sentence saying what heteroskedasticity breaks and one saying what it leaves alone. Then compare with your neighbour.

SECTION 2

Serial Correlation

1. Heteroskedasticity
2. Serial Correlation
3. Misspecification

Correlated Errors Have the Same Consequences as Heteroskedasticity

The goal of this section: put correlated errors in the same place as heteroskedasticity, and then argue that the correction is usually the wrong response.

When errors are correlated with each other — adjacent quarters, workers in the same firm, people in the same town — the estimates are still centred on the truth and the usual standard error is still wrong. Row two of the table is row one again.

Structurally there is nothing new here. What is new is what you should do about it, and it is not the same thing.

What a Run of Same-Signed Residuals Indicates

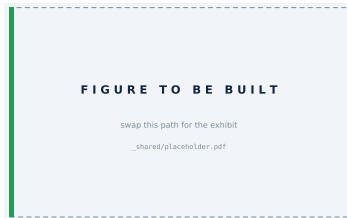


Figure 3: Residuals in order of schooling. They are negative at both ends and positive through the middle — not scattered, but arranged in runs.

Notes

Order these residuals by anything and you can compute a correlation. Here the pattern is not noise that happens to be correlated. It is the shape of something the model does not contain, showing through.

Why Correcting the Standard Error Comes Last

A correction exists: Newey & West (1987) gives a standard error valid under both heteroskedasticity and autocorrelation, and it is one argument in your regression call.

Why reaching for it first is the mistake

The correction widens the interval around a number you have not yet asked whether you should be reporting. Granger & Newbold (1974) is the case that settled this: two unrelated series, a large t -statistic and residuals that were badly correlated. The residuals were the warning. A wider interval around a meaningless coefficient is still meaningless.

Correlated residuals are a symptom. Ask what is missing, or what shape is wrong, before you ask which standard error to report.

Vote: What Would Cause This Pattern?

Residuals negative at both ends of the schooling range and positive through the middle.

- The errors are genuinely correlated with each other.
- A variable that matters is missing from the model.
- The relationship is curved and we fitted a straight line.
- You cannot tell from this plot which of the three it is.

Notes

Vote, argue with your neighbour, vote again. Then: what evidence would settle it?

Pause

Notes

Two minutes. Name two different faults that would produce the residual plot on the board. Then compare with your neighbour.

SECTION 3

Misspecification

1. Heteroskedasticity
2. Serial Correlation
3. Misspecification

Why the Wrong Functional Form Costs Unbiasedness

The goal of this section: show that getting the shape wrong is a different kind of failure from the first two, and a worse one.

Returns to schooling diminish. The truth bends, and we fitted a line. The line is then too low in the middle of the range and too high at the ends, which means the error does not average zero at every level of schooling.

That is assumption two, the one week 6 said buys unbiasedness. Break it and the estimate is no longer centred on anything in particular. No standard error repairs that, because there is nothing wrong with the standard error.

Curvature and Outliers in the Residual Plot

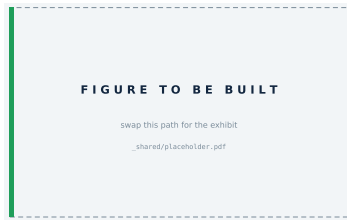


Figure 4: Left: residuals against fitted values with curvature switched on — an arch, not a cloud. Right: the same plot with a few very high earners in the sample — a cloud with four points a long way above it.

The point that summary statistics cannot be trusted to reveal a shape is Anscombe (1973), and it is fifty years old and still the reason the first thing you do with a fitted model is plot it.

Notes

Two plots, two faults, two different responses. The arch says the model has the wrong form. The four points say a handful of people are unlike the rest, and neither a robust standard error nor a quadratic term is the answer to that.

What the RESET Test Can and Cannot Tell You

There are named tests for this. Ramsey (1969) is the standard one, and it adds powers of the fitted values to the regression and asks whether they help.

It will reject for a missing variable, for the wrong functional form, for a handful of outliers, and for a sample large enough that any of the three is detectable. A rejection is a prompt to look at the plot. It is not a diagnosis, and it is certainly not a repair.

Run the plot first. If you run the test first you will spend the afternoon choosing a correction for a fault you have not identified.

Ranking the Three Faults by What They Cost

The claim to take away

A wrong standard error costs you an honest interval around the right number. A wrong shape costs you the number. Almost everyone worries more about the first, because it is the one with a test attached.

King & Roberts (2015) make a related argument that is worth holding on to: when the robust and the classical standard errors differ a lot, that gap is information about the model, not just a nuisance to be corrected away. Something in the specification is not doing its job.

Notes

This is why the third row of the table is set apart, and why 4.2 exists.

Vote: Does This Residual Plot Worry You?

A cloud of residuals against fitted values, plus four points sitting far above the rest — the four highest earners in the sample.

- Yes, and I would use a robust standard error.
- Yes, and I would look at who those four people are.
- Yes, and I would drop them.
- No, four points out of a hundred and twenty is nothing.

Notes

Vote, argue with your neighbour, vote again. Then: what evidence would settle it?

What Would Have To Be True?

The standing question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

[Candidate this week: a robust standard error makes the interval honest about noise. It says nothing about whether the line is the right shape, and nothing at all about whether schooling is what moved earnings.]

Application

Take one sample. Fit the usual line. Report the classical and the robust standard error side by side, then turn the fan on and do it again.

Notes

Vote first: as the error fans out more steeply, does the estimate move, does the classical standard error move, does the robust one, or some combination? Take the vote before anything is run.

Lab this week: turn on curvature instead and watch the same two standard errors both stay narrow while the estimate walks away from the truth.

What Each Fault Costs You

Today was three separate ideas, sorted by what each one takes away:

1 Heteroskedasticity.

The estimates survive and the standard errors do not, which is why robust is the default rather than a repair.

2 Serial Correlation.

The same damage with a different cause, and the correction is usually the wrong first move.

3 Misspecification.

This one takes unbiasedness, and no standard error can give it back.

Where 4.1 stops and 4.2 begins

Everything today was a fault you can see in the residuals and respond to. Next week is the faults no diagnostic will ever reveal, because the residual plot looks perfectly fine.

Before Lecture 4.2

- **Read** [chapter].
- **Do** the appendix derivation. It is the one that explains why the standard error in every piece of output you have read this term is the wrong one whenever the fan is on.
- **Try** the app: dial the fan up and watch the two standard errors come apart while the estimate stays put.

Problem set 4

[Out this week, due week n. Covers 4.1, and carries one question back from problem set 3.]

Thank you

sam.deegan@ucdconnect.ie
sam-deegan.com



BACKUP

Appendix

Derivation: Why Heteroskedasticity Breaks the Variance Formula

Take the week 6 decomposition of the slope into the truth plus a weighted sum of errors, and compute its variance without assuming the errors share a variance.

Notes

Three steps. Write the slope as the truth plus a sum of weights times errors. Take the variance of that sum, which is the sum of the squared weights times the individual error variances. Then stop, because there is no common σ^2 to pull out. Week 6 pulled it out in one line and never said that was a step; this is that step, refused. The result is the first formula on the slide, and the second formula is what you get by putting $\hat{u}_i^2$ where σ_i^2 stands. Why that substitution works is the interesting part: you cannot estimate n variances from n observations, but you never needed to — only one weighted sum of them.

[Full derivation. Set as homework; worked solution published with the set.]

Notes and Tools

- `plot(model, which = 1)`: residuals against fitted values, which is the first thing to look at and the last thing anyone does.
- `sandwich::vcovHC()` with `lmtest::coeftest()`, documented in [1](#). Its default is HC3.
- `estimatr::lm_robust()`, which reports HC2 with an adjusted degrees of freedom out of the box.
- `fixest::feols()` with `vcov = "hetero"`, if you are already using it.
- Stata's `vce(robust)` is HC1, which is not the same choice; say which one you used.

References

- Angrist, J. D., & Pischke, J.-S. (2017). Undergraduate econometrics instruction: Through our classes, darkly. *Journal of Economic Perspectives*, 31(2), 125–144. <https://doi.org/10.1257/jep.31.2.125>
- Anscombe, F. J. (1973). Graphs in statistical analysis. *The American Statistician*, 27(1), 17–21. <https://doi.org/10.1080/00031305.1973.10478966>
- Eicker, F. (1967). Limit theorems for regressions with unequal and dependent errors. In L. M. Le Cam & J. Neyman (Eds.), *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability, volume 1: statistics* (pp. 59–82). University of California Press.
- Granger, C. W. J., & Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120. [https://doi.org/10.1016/0304-4076\(74\)90034-7](https://doi.org/10.1016/0304-4076(74)90034-7)
- Huber, P. J. (1967). The behavior of maximum likelihood estimates under nonstandard conditions. In L. M. Le Cam & J. Neyman (Eds.), *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability, volume 1: statistics* (pp. 221–233). University of California Press.
- Imbens, G. W., & Kolesár, M. (2016). Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*, 98(4), 701–712. https://doi.org/10.1162/REST_a_00552

References

- King, G., & Roberts, M. E. (2015). How robust standard errors expose methodological problems they do not fix, and what to do about it. *Political Analysis*, 23(2), 159–179. <https://doi.org/10.1093/pan/mpu015>
- Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224. <https://doi.org/10.1080/00031305.2000.10474549>
- MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325. [https://doi.org/10.1016/0304-4076\(85\)90158-7](https://doi.org/10.1016/0304-4076(85)90158-7)
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. <https://doi.org/10.2307/1913610>
- Ramsey, J. B. (1969). Tests for specification errors in classical linear least-squares regression analysis. *Journal of the Royal Statistical Society: Series B (Methodological)*, 31(2), 350–371. <https://doi.org/10.1111/j.2517-6161.1969.tb00796.x>
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838. <https://doi.org/10.2307/1912934>